

Question Bank and PYQs

Chapter 1 - Sets

2 Mark Questions

Q1: Define a 'Set' according to the NCERT curriculum and provide one example. (2 Marks)

- A set is a well-defined collection of objects.
- Well-defined means that given an object, we can definitely determine if it belongs to the collection or not.
- Example: The collection of all vowels in the English alphabet is a set, denoted as $V = a, e, i, o, u$.
- The objects in a set are called its elements or members.

Q2: Write the set $A = x : x$ is an integer and $-3 < x < 7$ in roster form. (2 Marks)

- Identify the integers between -3 and 7 (excluding -3 and 7).
- The integers satisfying this condition are $-2, -1, 0, 1, 2, 3, 4, 5, 6$.
- Write these elements separated by commas within braces.
- The roster form is $A = -2, -1, 0, 1, 2, 3, 4, 5, 6$.

Q3: Explain the concept of an Empty Set (Null Set) with an example. (2 Marks)

- A set which does not contain any element is called the empty set or the null set or the void set.
- It is denoted by the symbol ϕ or empty braces $\{\}$.
- Example: The set $A = x : x$ is a student currently studying in both Class 10 and Class 11.
- Since no student can be in both classes simultaneously, $A = \phi$.

Q4: Differentiate between finite and infinite sets. (2 Marks)

- A set which is empty or consists of a definite number of elements is called a finite set.
- A set where the process of counting elements does not come to an end is called an infinite set.

- Example of finite: Set of days in a week.
- Example of infinite: The set of all natural numbers $N = 1, 2, 3, \dots$

Q5: When are two sets A and B said to be equal? (2 Marks)

- Two sets A and B are said to be equal if they have exactly the same elements.
- Mathematically, we write $A = B$.
- If they are not equal, we write $A \neq B$.
- The order in which elements are listed does not change the equality of sets.

Q7: Define a Power Set and find $P(A)$ if $A = a, b$. (2 Marks)

- The collection of all subsets of a set A is called the power set of A .
- It is denoted by $P(A)$.
- For $A = a, b$, the subsets are ϕ, a, b, a, b .
- Therefore, $P(A) = \phi, a, b, a, b$.

Q8: What is a Universal Set? Give an example. (2 Marks)

- A set that contains all objects or elements under consideration in a particular context is called the Universal Set.
- It is usually denoted by U .
- Example: For the set of all right triangles, the Universal set U could be the set of all triangles in a plane.
- All other sets in that context are subsets of U .

Q9: Represent the following as intervals: $x : x \in R, -4 < x \leq 6$. (2 Marks)

- Check the inequality signs to determine the type of bracket.
- The symbol $<$ indicates an open interval (round bracket '(').
- The symbol $\leq$ indicates a closed interval (square bracket ']').
- The interval representation is $(-4, 6]$.

Q10: State whether the following is true or false: $a, b \subseteq a, b, c$. Justify. (2 Marks)

- True.

- A set A is a subset of B if every element of A is also an element of B .
- Elements a and b are present in both sets.
- Therefore, the condition for a subset is satisfied.

3 Mark Questions

Q6: Write all the subsets of the set $A = 1, 2, 3$. (3 Marks)

- Identify the number of elements $n = 3$. Total subsets $= 2^n = 2^3 = 8$.
- The subsets are: ϕ (the empty set).
- Single-element sets: 1, 2, 3.
- Two-element sets: 1, 2, 1, 3, 2, 3.
- The set itself: 1, 2, 3.

Q11: If $A = 2, 4, 6, 8$ and $B = 6, 8, 10, 12$, find $A \cup B$ and $A \cap B$. (3 Marks)

- To find $A \cup B$, list all elements that are in A , or in B , or in both.
- $A \cup B = 2, 4, 6, 8, 10, 12$.
- To find $A \cap B$, list only the elements that are common to both A and B .
- The common elements are 6 and 8.
- $A \cap B = 6, 8$.

Q12: Let $A = 1, 2, 3, 4, 5, 6$, $B = 2, 4, 6, 8$. Find $A - B$ and $B - A$. (3 Marks)

- The difference $A - B$ consists of elements that belong to A but not to B .
- Elements of A not in B are 1, 3, 5. So, $A - B = 1, 3, 5$.
- The difference $B - A$ consists of elements that belong to B but not to A .
- Element of B not in A is 8. So, $B - A = 8$.
- Note that $A - B \neq B - A$.

Q13: Define Disjoint Sets with an example and represent them using a Venn diagram description. (3 Marks)

- Two sets A and B are said to be disjoint if $A \cap B = \phi$.
- This means they have no common elements.
- Example: $A = 1, 3, 5$ and $B = 2, 4, 6$. Here $A \cap B = \phi$.
- In a Venn diagram, disjoint sets are represented by two circles that do not intersect or overlap.

Q14: Let $U = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$ be the universal set and $A = 1, 3, 5, 7, 9$. Find A' and show that $(A')' = A$. (3 Marks)

- The complement A' is the set of elements in U that are not in A .
- $A' = 2, 4, 6, 8, 10$.
- The complement of A' , denoted $(A')'$, is the set of elements in U not in A' .
- Comparing A' with U , the elements not in A' are $1, 3, 5, 7, 9$.
- Thus, $(A')' = 1, 3, 5, 7, 9 = A$.

Q15: If X and Y are two sets such that $X \cup Y$ has 50 elements, X has 28 elements and Y has 32 elements, how many elements does $X \cap Y$ have? (3 Marks)

- Given: $n(X \cup Y) = 50$, $n(X) = 28$, $n(Y) = 32$.
- Use the formula: $n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$.
- Substitute values: $50 = 28 + 32 - n(X \cap Y)$.
- $50 = 60 - n(X \cap Y)$.
- $n(X \cap Y) = 60 - 50 = 10$.
- So, $X \cap Y$ has 10 elements.

Q16: Show that the set of letters needed to spell 'CATARACT' and the set of letters needed to spell 'TRACT' are equal. (3 Marks)

- Let L_1 be the set of letters in 'CATARACT'.
- In roster form, $L_1 = C, A, T, R$ (distinct elements only).
- Let L_2 be the set of letters in 'TRACT'.
- In roster form, $L_2 = T, R, A, C$.

- Since L_1 and L_2 contain exactly the same elements, $L_1 = L_2$.

Q17: List all the subsets of the set $A = \phi, \phi$. (3 Marks)

- The set A has two elements: $e_1 = \phi$ and $e_2 = \phi$.
- Number of subsets $= 2^2 = 4$.
- The subsets are: ϕ (the empty set which is a subset of every set).
- ϕ (set containing the first element).
- ϕ (set containing the second element).
- $\phi, \phi(\text{thesetitself})$.

Q18: State and verify the Commutative law for Union of sets using $A = 1, 2$ and $B = 2, 3$. (3 Marks)

- Commutative Law: $A \cup B = B \cup A$.
- Find $A \cup B$: Elements in A or B are 1, 2, 3. So, $A \cup B = 1, 2, 3$.
- Find $B \cup A$: Elements in B or A are 2, 3, 1. So, $B \cup A = 1, 2, 3$.
- Since $A \cup B = B \cup A$, the law is verified.

Q19: Draw a Venn diagram to illustrate $A \cap (B \cup C)$. (3 Marks)

- Draw three intersecting circles representing sets A , B , and C inside a rectangle (Universal set U).
- Identify the region representing $B \cup C$ (the entire area covered by circles B and C).
- Identify the region of set A .
- Shade the part of circle A that overlaps with the combined area of B and C .
- This shaded region represents $A \cap (B \cup C)$.

Q20: If $A \subseteq B$, then show that $A \cup B = B$. (3 Marks)

- By definition, $A \cup B$ contains all elements of A and all elements of B .
- Since $A \subseteq B$, every element of A is already an element of B .
- Therefore, the collection of all elements in A and B is simply the collection of elements in B .

- Thus, $A \cup B = B$.

5 Mark Questions

Q21: In a survey of 600 students in a school, 150 students were found to be taking tea and 225 taking coffee, 100 were taking both tea and coffee. Find how many students were taking neither tea nor coffee. (5 Marks)

- Let U be the set of all students surveyed. $n(U) = 600$.
- Let T be the set of students taking tea. $n(T) = 150$.
- Let C be the set of students taking coffee. $n(C) = 225$.
- Number of students taking both tea and coffee is $n(T \cap C) = 100$.
- Calculate students taking either tea or coffee using the formula: $n(T \cup C) = n(T) + n(C) - n(T \cap C)$.
- $n(T \cup C) = 150 + 225 - 100 = 275$.
- The students taking neither tea nor coffee are those in the complement of the union: $(T \cup C)'$.
- Number of students taking neither = $n(U) - n(T \cup C)$.
- $n(\text{neither}) = 600 - 275 = 325$.
- Final Answer: 325 students were taking neither tea nor coffee.

Q22: Prove De Morgan's Law: $(A \cup B)' = A' \cap B'$ for any two sets A and B . (5 Marks)

- Part 1: To show $(A \cup B)' \subseteq A' \cap B'$. Let $x \in (A \cup B)'$.
- Then $x \notin (A \cup B) \implies x \notin A$ and $x \notin B$.
- This implies $x \in A'$ and $x \in B' \implies x \in A' \cap B'$. Thus, $(A \cup B)' \subseteq A' \cap B'$.
- Part 2: To show $A' \cap B' \subseteq (A \cup B)'$. Let $y \in A' \cap B'$.
- Then $y \in A'$ and $y \in B' \implies y \notin A$ and $y \notin B$.
- This implies $y \notin (A \cup B) \implies y \in (A \cup B)'$. Thus, $A' \cap B' \subseteq (A \cup B)'$.
- Conclusion: From Part 1 and Part 2, $(A \cup B)' = A' \cap B'$.

- Mention: This law can also be verified visually using a Venn diagram by shading the exterior of the union circle.

Q23: In a group of 65 people, 40 like cricket, 10 like both cricket and tennis. How many like tennis only and not cricket? How many like tennis? (5 Marks)

- Let C be the set of people who like cricket and T be the set of people who like tennis.
- Given: $n(C \cup T) = 65$, $n(C) = 40$, $n(C \cap T) = 10$.
- Step 1: Find $n(T)$ using the formula $n(C \cup T) = n(C) + n(T) - n(C \cap T)$.
- $65 = 40 + n(T) - 10 \implies 65 = 30 + n(T) \implies n(T) = 35$. So, 35 people like tennis.
- Step 2: Find people who like tennis only ($T - C$).
- Use the formula: $n(T - C) = n(T) - n(C \cap T)$.
- $n(T - C) = 35 - 10 = 25$.
- Alternative check: $n(C \cup T) = n(C - T) + n(T - C) + n(C \cap T)$.
- $n(C - T) = n(C) - n(C \cap T) = 40 - 10 = 30$.
- Verification: $30 + 25 + 10 = 65$ (Total matches). Final answers: 25 like tennis only, 35 like tennis.

Q24: Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ using the property of elements. (5 Marks)

- This is the Distributive Law of Intersection over Union.
- Part 1: Let $x \in A \cap (B \cup C)$.
- This means $x \in A$ AND $x \in (B \cup C)$.
- This implies $x \in A$ AND ($x \in B$ OR $x \in C$).
- Case i: $x \in A$ and $x \in B \implies x \in (A \cap B)$. Case ii: $x \in A$ and $x \in C \implies x \in (A \cap C)$.
- In either case, $x \in (A \cap B) \cup (A \cap C)$. Thus $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.
- Part 2: Let $y \in (A \cap B) \cup (A \cap C)$.
- Then $y \in (A \cap B)$ OR $y \in (A \cap C)$.
- If $y \in (A \cap B)$, then $y \in A$ and $y \in B$. If $y \in (A \cap C)$, then $y \in A$ and $y \in C$.
- In both cases, $y \in A$ and $y \in (B \cup C)$. Thus $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

- Combining both parts, the equality is proved.

Q25: In a survey of 25 students, it was found that 15 had taken Mathematics, 12 had taken Physics and 11 had taken Chemistry, 5 had taken Mathematics and Chemistry, 9 had taken Mathematics and Physics, 4 had taken Physics and Chemistry and 3 had taken all the three subjects. Find the number of students that had taken: (i) only Chemistry (ii) none of the subjects. (5 Marks)

- Let M, P, C be the sets of students taking Math, Physics, and Chemistry respectively.
- Given: $n(M) = 15, n(P) = 12, n(C) = 11, n(M \cap C) = 5, n(M \cap P) = 9, n(P \cap C) = 4, n(M \cap P \cap C) = 3$.
- Step 1: Find students taking ONLY Chemistry. Formula: $n(C_{\text{only}}) = n(C) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$.
- $n(C_{\text{only}}) = 11 - 5 - 4 + 3 = 5$.
- Step 2: Find total students taking at least one subject: $n(M \cup P \cup C)$.
- $n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$.
- $n(M \cup P \cup C) = 15 + 12 + 11 - 9 - 5 - 4 + 3 = 23$.
- Step 3: Students taking none of the subjects $= n(U) - n(M \cup P \cup C)$.
- $n(\text{none}) = 25 - 23 = 2$.
- Final Answers: (i) 5 (ii) 2.

Q26: Explain the concept of intervals as subsets of R . Differentiate between open, closed, semi-open, and semi-closed intervals with mathematical notation and number line descriptions. (5 Marks)

- An interval is a subset of real numbers such that any number lying between two numbers in the set is also included in the set.
- Open Interval: $(a, b) = x : a < x < b$. The endpoints a and b are excluded. On a number line, it is represented by open circles at a and b .
- Closed Interval: $[a, b] = x : a \leq x \leq b$. The endpoints a and b are included. On a number line, it is represented by dark solid circles at a and b .

- Semi-open/Semi-closed: $[a, b) = \{x : a \leq x < b\}$ (includes a , excludes b) and $(a, b] = \{x : a < x \leq b\}$ (excludes a , includes b).
- Infinity in intervals: The set of non-negative real numbers is $[0, \infty)$, where infinity always uses a round bracket because it is not a reachable real number.
- Intervals allow us to represent continuous sets of numbers which cannot be listed in roster form.

Q27: For any two sets A and B , prove that $P(A \cap B) = P(A) \cap P(B)$. (5 Marks)

- Let X be an arbitrary element in $P(A \cap B)$.
- By definition of power set, $X \subseteq (A \cap B)$.
- This implies $X \subseteq A$ AND $X \subseteq B$.
- Since $X \subseteq A$, then $X \in P(A)$. Since $X \subseteq B$, then $X \in P(B)$.
- Therefore, $X \in P(A) \cap P(B)$, which shows $P(A \cap B) \subseteq P(A) \cap P(B)$.
- Now, let $Y \in P(A) \cap P(B)$. Then $Y \in P(A)$ and $Y \in P(B)$.
- This implies $Y \subseteq A$ and $Y \subseteq B$, which means $Y \subseteq (A \cap B)$.
- By definition of power set, $Y \in P(A \cap B)$, which shows $P(A) \cap P(B) \subseteq P(A \cap B)$.
- From both directions, we conclude $P(A \cap B) = P(A) \cap P(B)$.

Q28: Let A, B and C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$. Show that $B = C$. (5 Marks)

- Start with $B = B \cap (B \cup A)$ [Absorption Law].
- Substitute $A \cup B$ with $A \cup C$: $B = B \cap (A \cup C)$.
- Apply distributive law: $B = (B \cap A) \cup (B \cap C)$.
- Substitute $A \cap B$ with $A \cap C$: $B = (A \cap C) \cup (B \cap C)$.
- Now start with C : $C = C \cap (C \cup A)$.
- Substitute $A \cup C$ with $A \cup B$: $C = C \cap (A \cup B)$.
- Apply distributive law: $C = (C \cap A) \cup (C \cap B)$.

- Notice that $(A \cap C) \cup (B \cap C)$ is the same as $(C \cap A) \cup (C \cap B)$ because intersection and union are commutative.
- Therefore, $B = C$.

Q29: A college awarded 38 medals in football, 15 in basketball and 20 in cricket. If these medals went to a total of 58 men and only 3 men got medals in all the three sports, how many received medals in exactly two of the three sports? (5 Marks)

- Let F, B, C be the sets of men receiving medals in Football, Basketball, and Cricket.
- Given: $n(F) = 38, n(B) = 15, n(C) = 20, n(F \cup B \cup C) = 58, n(F \cap B \cap C) = 3$.
- Use the formula: $n(F \cup B \cup C) = n(F) + n(B) + n(C) - [n(F \cap B) + n(B \cap C) + n(C \cap F)] + n(F \cap B \cap C)$.
- $58 = 38 + 15 + 20 - [n(F \cap B) + n(B \cap C) + n(C \cap F)] + 3$.
- $58 = 76 - [n(F \cap B) + n(B \cap C) + n(C \cap F)]$.
- Let $S = n(F \cap B) + n(B \cap C) + n(C \cap F) = 76 - 58 = 18$.
- The number of people who received medals in exactly two sports is given by $S - 3 \times n(F \cap B \cap C)$.
- Wait, let's use the regions: Let a, b, c be people in exactly two sports. $S = (a + 3) + (b + 3) + (c + 3)$ is not correct. The correct logic is $S = a + b + c + 3[n(F \cap B \cap C)]$.
- So, exactly two sports $= S - 3[n(F \cap B \cap C)] = 18 - 3(3) = 18 - 9 = 9$.
- Final Answer: 9 men received medals in exactly two of the three sports.

Q30: Prove that for any sets A and B , $A = (A \cap B) \cup (A - B)$ and $A \cup (B - A) = A \cup B$. (5 Marks)

- Part 1: To show $A = (A \cap B) \cup (A - B)$.
- By definition, $A - B = A \cap B'$.
- So, $(A \cap B) \cup (A - B) = (A \cap B) \cup (A \cap B')$.
- By Distributive Law: $A \cap (B \cup B')$.
- Since $B \cup B' = U$ (Universal set), the expression becomes $A \cap U = A$. Hence proved.
- Part 2: To show $A \cup (B - A) = A \cup B$.

- Write $B - A$ as $B \cap A'$.
- The expression becomes $A \cup (B \cap A')$.
- By Distributive Law: $(A \cup B) \cap (A \cup A')$.
- Since $A \cup A' = U$, the expression becomes $(A \cup B) \cap U = A \cup B$. Hence proved.

